Rally also supports YAML files to run jobs. The usage is as follows (consider *task.yaml*):

```
rally task start task.yaml
```

Rally scenarios in general can be found at: *https://rally.readthedocs.io/en/latest/plugins/plugin_reference.html*

Rally jobs for Octavia can be found at *https://opendev.org/openstack/rally-openstack/src/branch/master/samples/tasks/scenarios/octavia.*

In case a load balancer is to be created using OVN instead of Amphora, an additional key-value pair is added in the *args* section to specify the provider:

```
"args": {
    "provider": "ovn"
}
```

Tests are conducted using the scenarios given below.

Octavia.create_and_delete_loadbalancers: Creates a load balancer for each subnet and then Figures 1 and 2 show the results of this scenario.

Octavia.create_and_list_loadbalancers: Creates a loadbalancer per each subnet and then lists loadbalancers. Figures 3 and 4 show the results of this scenario.

Octavia.create_and_stats_loadbalancers: Creates a loadbalancer for each subnet and then compares these. Figures 5 and 6 show the results of this scenario.

Octavia.create_and_show_loadbalancers: Creates a loadbalancer per each subnet and stats. Figures 7 and 8 show the results of this scenario.

Octavia.create_and_update_loadbalancers: Creates a loadbalancer for each subnet and then updates them. Figures 9 and 10 show the results of this scenario.

In Figure 1, the 'Service-level agreement' table shows the failure rate of the *Octavia.create_and_delete_loadbalancers* scenario and the 'Total durations' table shows the total time taken for the job and for each of the sub-tasks, i.e., the sum of times taken for creating the load balancer in each iteration (the number of iterations is five for each of the jobs run), the sum of times taken waiting for the load balancer to create in each iteration, and the sum of times taken for deleting the load balancer in each iteration. The failure rate is 100 per cent, which means that none of the iterations of the job are completed successfully. Similarly, Figures 3, 5, 7 and 9 show the results of the jobs mentioned in each of them respectively when Amphora is used and Figures 2, 4, 6, 8, 10 show the results of the same when OVN is used. Figure 11 shows the errors which occurred when running the *Octavia.create_and_delete_loadbalancers* Rally job when Amphora is used. The same

Octavia.create_and_stats_loadbalancers (58.517s)

Create a loadbalancer per each subnet and stats

Overview Details Failures Input task

Load duration: **26.358 s** Full duration: **58.517 s** Iterations: **5** Failures: **5** Started at: **2020-11-20T07:41:37**

Service-level agreement

Criterion	Detail	Success
failure_rate	Failure rate criteria 0.00% <= 100.00% <= 0.00% - Failed	False

Total durations

Action	Min (sec)	Median (sec)	90%ile (sec)	95%ile (sec)	Max (sec)	Avg (sec)	Success	Count
octavia.load_balancer_create	1.339	2.392	2.728	2.768	2.808	2.28	100.0%	5
octavia.wait_for_loadbalancers	6.239	6.277	7.508	7.888	8.268	6.679	0.0%	5
-> octavia.load_balancer_show (x4)	*0.233*	*0.254*	*0.308*	*0.316*	*0.324*	*0.266*	*100.0%*	*4*
-> octavia.load_balancer_show (x5)	*0.262*	*0.262*	*0.262*	*0.262*	*0.262*	*0.262*	*100.0%*	*1*
total	**7.577**	**8.669**	**10.196**	**10.536**	**10.876**	**8.959**	**0.0%**	**5**
-> duration	*7.577*	*8.669*	*10.196*	*10.536*	*10.876*	*8.959*	*0.0%*	*5*
-> idle_duration	*0*	*0*	*0*	*0*	*0*	*0*	*0.0%*	*5*

Figure 7: Result of *Octavia.create_and_stats_loadbalancers* task using Amphora

Octavia.create_and_stats_loadbalancers (79.639s)

Create a loadbalancer per each subnet and stats

Overview Details Input task

Load duration: **20.374 s** Full duration: **79.639 s** Iterations: **5** Failures: **0** Started at: **2020-11-18T06:48:05**

Service-level agreement

Criterion	Detail	Success
failure_rate	Failure rate criteria 0.00% <= 0.00% <= 0.00% - Passed	True

Total durations

Action	Min (sec)	Median (sec)	90%ile (sec)	95%ile (sec)	Max (sec)	Avg (sec)	Success	Count
octavia.load_balancer_create	3.824	6.345	7.095	7.16	7.242	5.824	100.0%	5
octavia.wait_for_loadbalancers	0.049	1.784	2.428	2.476	2.525	1.417	100.0%	5
-> octavia.load_balancer_show (x2)	*0.28*	*0.388*	*0.475*	*0.486*	*0.497*	*0.388*	*100.0%*	*2*
-> octavia.load_balancer_show	*0.049*	*0.447*	*1.516*	*1.65*	*1.783*	*0.76*	*100.0%*	*3*
octavia.load_balancer_stats_show	0.037	0.058	0.272	0.342	0.412	0.122	100.0%	5
total	**3.91**	**7.033**	**9.528**	**9.558**	**9.587**	**7.364**	**100.0%**	**5**
-> duration	*3.91*	*7.033*	*9.528*	*9.558*	*9.587*	*7.364*	*100.0%*	*5*

Figure 8: Result of *Octavia.create_and_stats_loadbalancers* task using OVN